"A compass guided by past findings."

LITERATURE SEARCH

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Defining Literature Search

Literature Search is the process of *seeking out* and *identifying the existing and available literature* related to a topic or question of interest.

Conducting a literature search is a crucial step in gathering relevant information and building a solid foundation for your investigation.

IMPORTANCE OF LITERATURE SEARCH

- Gather relevant studies on your relevant topic
- Assess the current evidence base
- Identify appropriate methodologies
- Guide the development of research tools and interventions

MAJOR STEPS IN LITERAURE SEARCH

1. Create a well-defined research or topic question
2. Brainstorm to gather subject terms, keywords and synonyms
3. Develop the search strategy
4. Select database(s) to search
5. Evaluate Your Results
6. Organize Your Results

Create a well-defined research or topic question

The first step in a literature search is to construct a well-defined question.

This helps in ensuring a *comprehensive* and *efficient* search of the available literature for relevant publications on your topic.

The well-constructed research question provides guidance for determining ***search terms***.

A good or well-constructed research question is:

Original and of interest

clear and focused:

The question concept is researchable in terms of time and access to a suitable amount of quality research resources.

One well-established way that can be used both for creating research questions and developing strategies is known as PICO(T).

Asking for the required information in the form of a question

- **PICO**
- **Patient population:** type of patient
- **Intervention:** new approach or strategy of treatment
- **Comparison:** the control intervention
- **Outcome:** clinically meaningful outcomes that are important for the patients.
- **Patient** or **population** or **problem:** age, sex, race, disease severity and co-morbidity
- **Population of patient with a problem**
- **Intervention:** new intervention; independent or input variable
- **Comparison** or **comparator** or **control:** one intervention to which new intervention is compared
  - comparator is **"gold standard"**
- **Outcome:** health consequences of exposure or intervention both benefits and adverse effects

Brainstorm to gather subject terms, keywords and synonyms.

Now that we have developed a well-constructed research question, next up is creating a list of search terms.

Starting with your well-built research question, now create a list of search terms using elements from your question. It may not be necessary to include all of the question elements in your search, but key concepts are a good place to start.

Break the question down into its individual or key concepts.

Think broadly and abstractly!

Gather Search Terms & Synonyms

Decide on the words or phrases to describe the concepts**.**

Consider all possible term words, keywords or phrases that might be used to describe the concepts of your topic.

These should include synonyms, variations in spelling (US vs. UK/ Canadian), word endings (singular or plural), variant terminology, acronyms and related terms for each key term/concept. Truncation or “wild cards” and phrase searching are two techniques you may also find useful .

Controlled Vocabulary/Subject Headings/Subject Thesaurus.

Controlled vocabularies or thesauri are standardized, organized sets of terms that databases use to describe content in a consistent fashion.

Many databases have their own, unique controlled vocabulary. PubMed/Medline uses Medical Subject Headings (MeSH).

Here is a sample question: *What are the social and ethical implications of genetic testing for breast cancer susceptibility*?

Breaking down the question into its individual or key concepts could be:

Genetic testing

Breast cancer susceptibility and

Potential concepts for *ethical implications* could be: Health care providers "duty to warn" about harmful effects / Patient's responsibility to inform health care provider about family history / Confidentiality of health information / Potential for genetic discrimination / Implications of genetic responsibility

Develop the search strategy

Once you have collected all of the possible search terms and decided which databases to search, now it is time to construct your search strategy.

To construct your search strategy, combine controlled vocabulary terms, keywords and phrases using words such as "**OR" and "AND",** known as *Boolean operators*, to broaden or narrow your search results.

Boolean operators are powerful tools that are part of a system of logic designed to yield optimal search results.

Boolean operators act by establishing relationships between groups or when you combine sets of search terms and define the relationship between the individual sets.

Set parameters for your search.

- If needed, apply filters to narrow your search.
- A date limit may be helpful, involves more recent technology, theories or concepts (e.g. social media, telemedicine, smart phones, AI, etc.).
- publication type, such as review, randomized control trials, case studies, peer-reviewed, scholarly, etc.
- age, such as child, teen, adult, elderly.

Select database(s) to search

After building your search strategy, it is time to test it out.

Select core and specialized databases to search.

The databases selected will depend on the topic to be searched.

Databases Use

PubMed

Google Scholar

Scopus

Cochrane Library

Web of Science

Local/regional databases

Evaluate Your Results

After conducting your search, you'll be left with a large number of results. **Evaluate each article individually** to determine its relevance to your research:

Read the title, abstract, and conclusion

Check the publication date and impact factor.

Assess the article's methodology, findings, and conclusions.

Identify gaps in the literature and areas for further research

Organize Your Results

As you evaluate your results, **organize and synthesize your findings**:

- Create a spreadsheet or table to track your results
- Identify patterns, themes, and emerging trends
- Identify gaps in the literature and areas for further research
- Highlight the most relevant and influential studies

Use specific search terms to reduce irrelevant results

Use advanced search features to filter results by author, date, and publication type

Use citation management software to organize and format your citations

Use quotation marks to search for exact phrases

Keep track of your search process to ensure transparency and reproducibility

TAKE HOME MESSAGE

Remember to stay focused, organized, and patient –

a good literature search takes time and effort, but it’s essential for building a strong foundation for your research.